

FRONT-END WEB DEVELOPMENT

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1

TYPESCRIPT TYPES

INTRODUCTION

- **Primitive Types**

- Boolean
- String
- Number
- Array
- Enum
- Any
- Void

BOOLEAN

- The most basic datatype is the simple true/false value.
 - **let** isFalse: **boolean** = **false**;

STRING

- String types plays a very vital role in any language, as in other languages, we use the type 'string' to refer to these textual datatypes. TypeScript uses the double quote (") or single quote (') to surround string data.
 - **let** name: string = "abc";

NUMBER

- As in JavaScript, all numbers in TypeScript are floating point values. These 'floating point' numbers get the type 'number'.
- **let** height: number = 99;

ARRAY

- TypeScript allows you to work with arrays of values. Array types can be written in one of two ways. In the first method, you use the type of the elements followed by '[]' to denote an array of that element type.

```
let list:number[] = [1, 2, 3];
```

- The second method uses a generic array type 'Array<elemType>'

```
let list:Array<number> = [1, 2, 3];
```

ENUM

- An 'enum' is an easy way of giving friendly names to sets of numeric values.
 - `enum Color {pen = 1, pencil = 2, eraser = 4};`
 - `console.log(Color.eraser)`
- By default, enums begin numbering their members starting at 0. You can change this by manually setting the value of one its members.
 - **enum** `stuff {pen, pencil, eraser};`
 - `console.log(stuff.eraser)`

ANY

- We may need to describe the type of variables that we may not know when we are writing the application. These values may come from dynamic content:
 - `let notSure: any = 4;`
 - `notSure = "maybe a string instead";`
 - `notSure = false; // okay, definitely a boolean`

VOID

- The opposite in some ways to 'any' is 'void', the absence of having any type at all. You may commonly see this as the return type of functions that do not return a value.
 - **function** warnUser(): **void** {
 - alert("This is my warning message");
 - }

FUNCTION TAKING ARGUMENT (TAKES A STRING ARG — RETURNS A NUMBER)

- `function myfunction(value: string): number {`
- `let num: number = value.length;`
- `return num;`
- `}`

LAST SLIDE